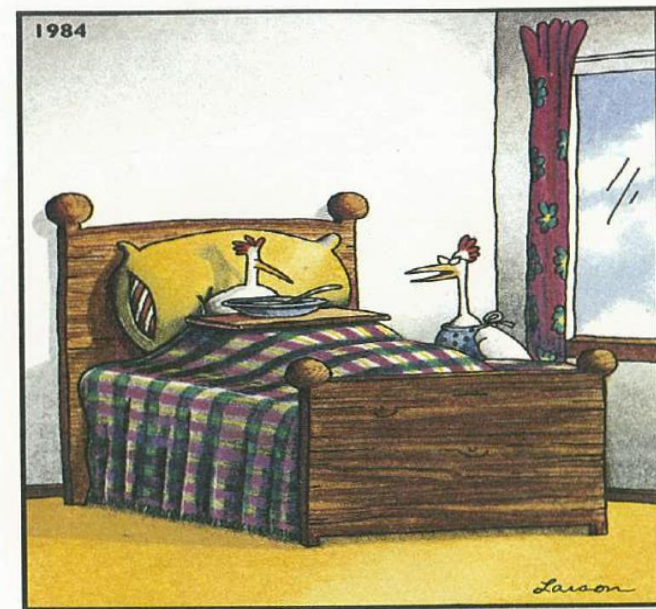
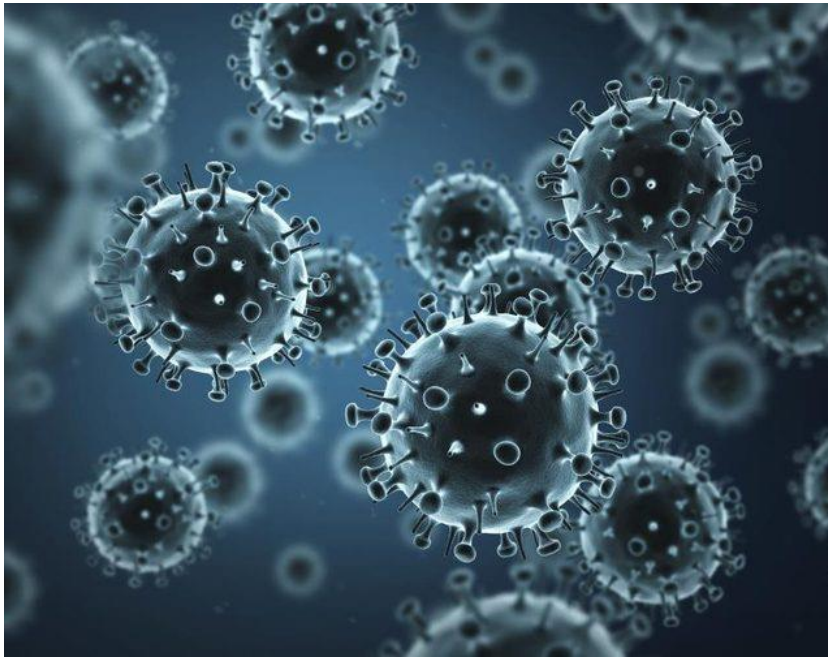
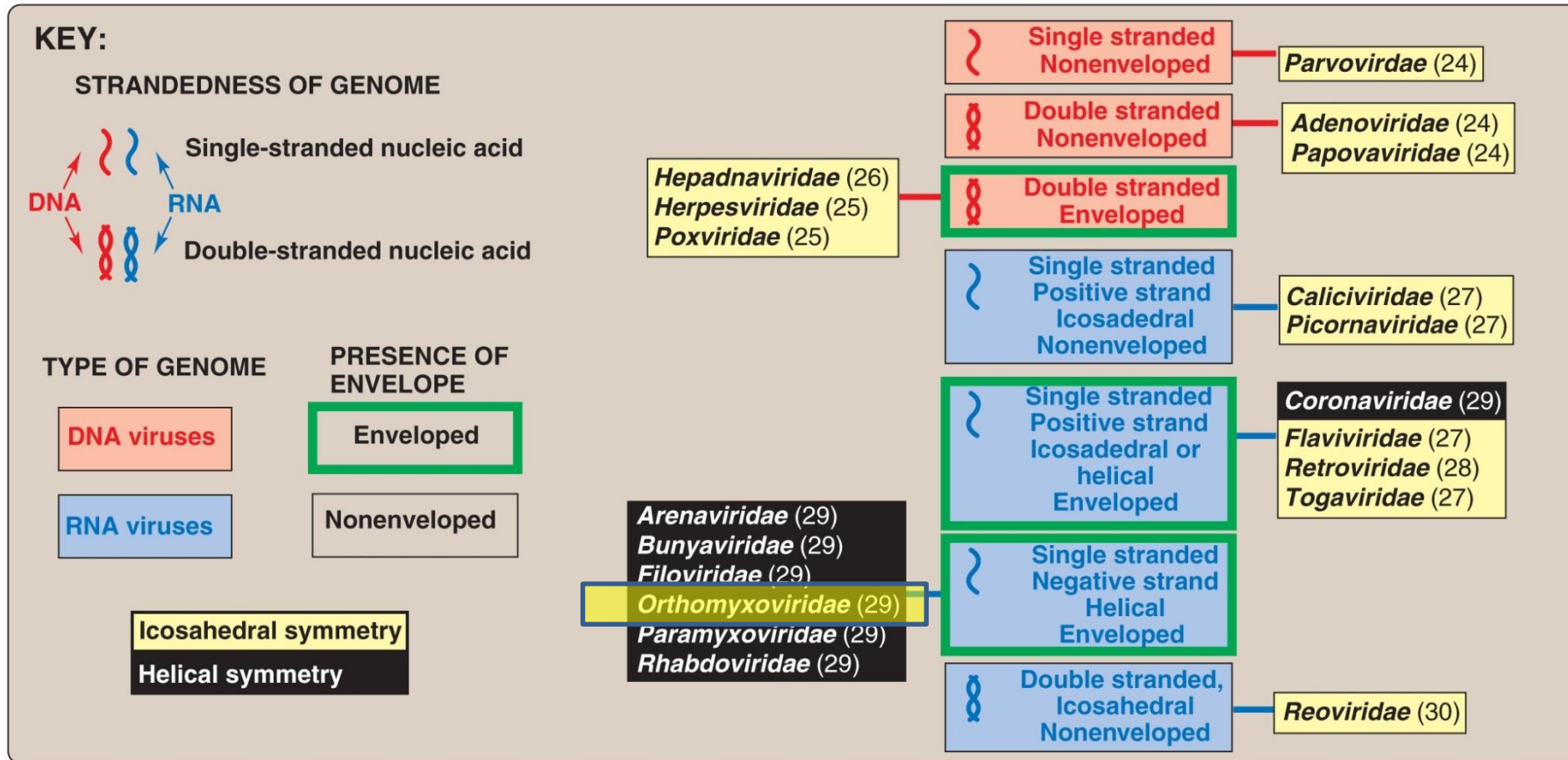


Influenza Virus



"Quit complaining and eat it! ... Number one, chicken soup is good for the flu—and number two, it's nobody we know."

Viruses

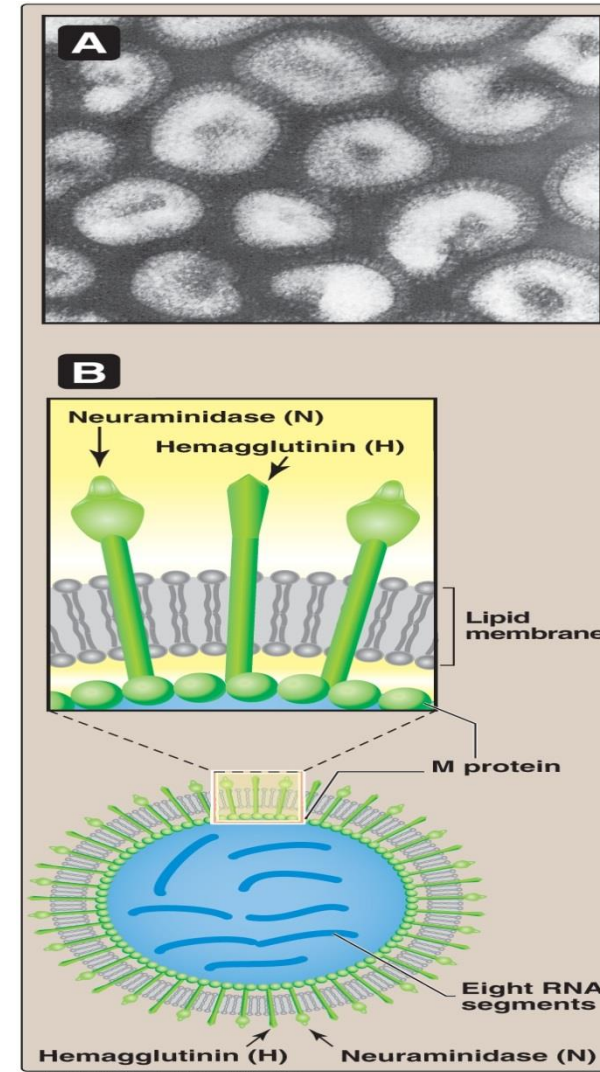


Copyright © 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins

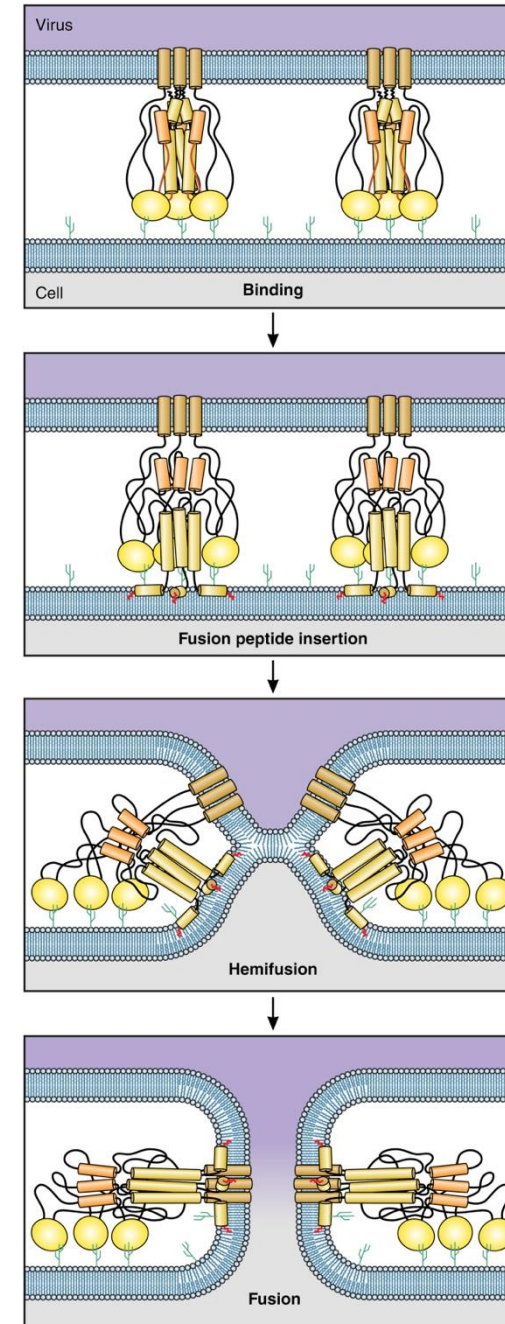
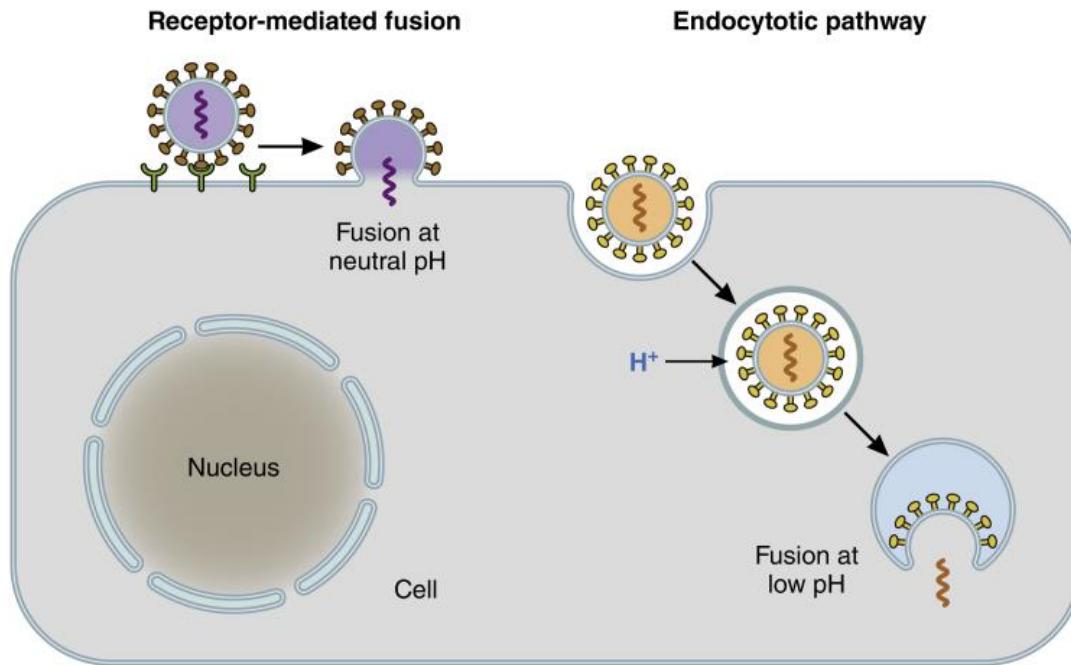
Today we will be going over
Influenza

Influenza Virus

- Orthomixoviridae
- 3 coat proteins (type A)
 - N (Neuraminidase)
 - H (Hemagglutinin)
 - M protein
- 4 coat proteins (type B)
- H, N, NB, and M2



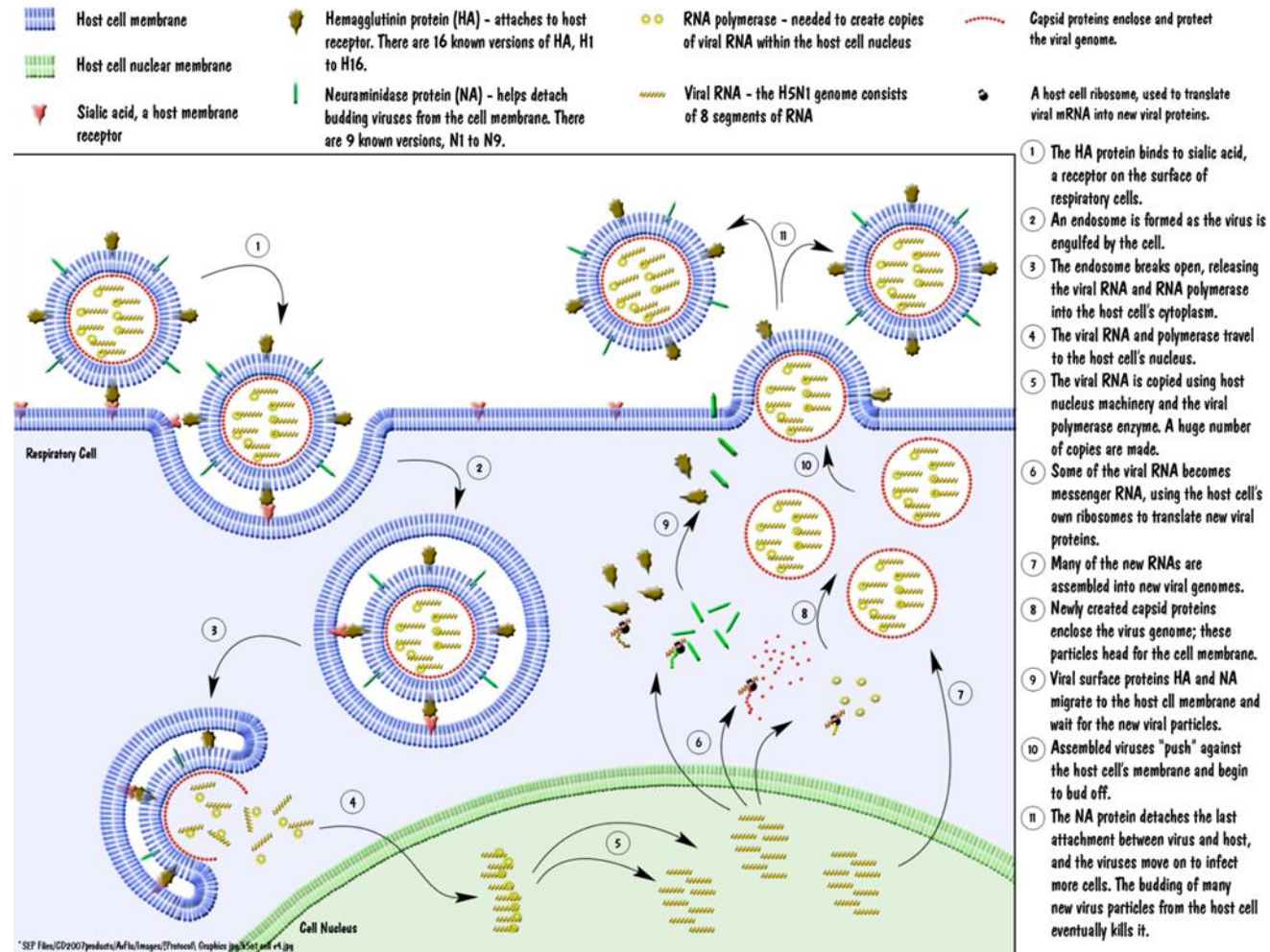
Virus entry of a cell



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4788752/>
<https://www.nature.com/articles/nrmicro817>

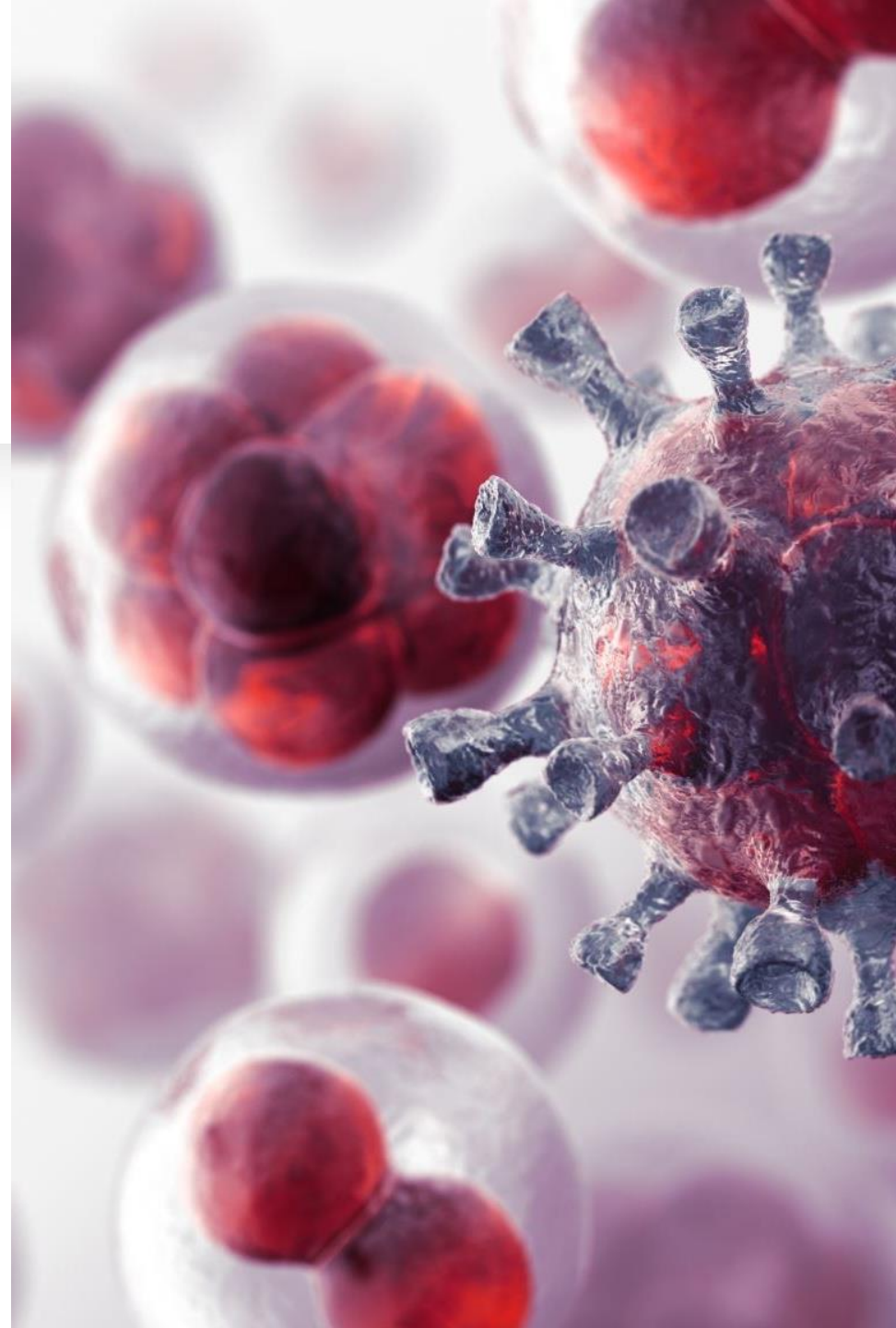
Virus infection

- How does the virus infect a host cell?

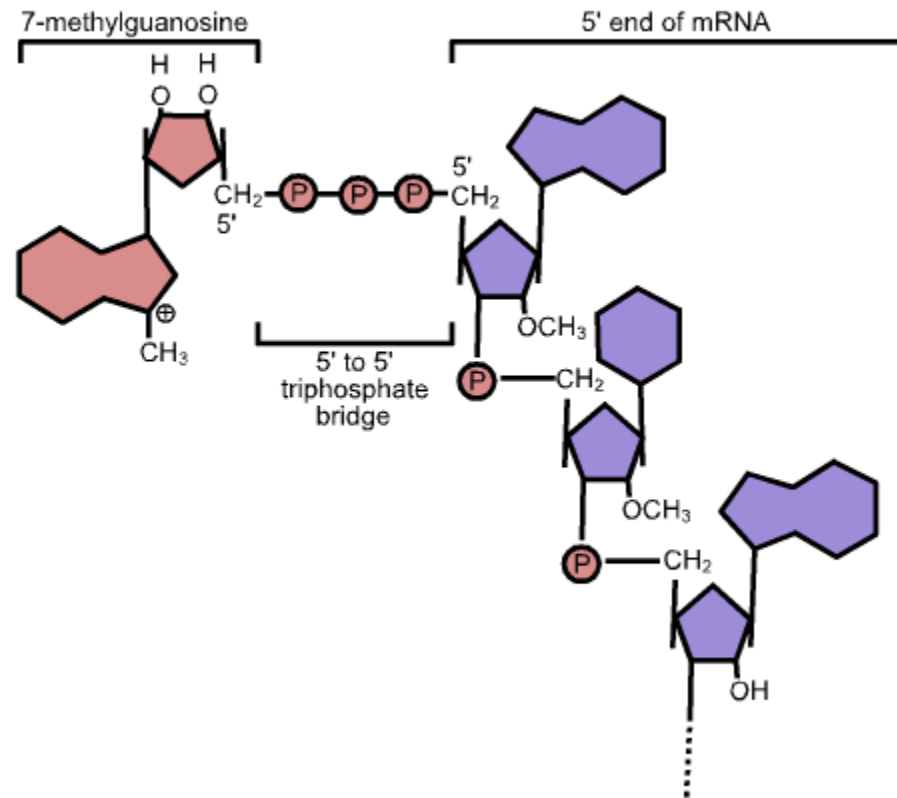


Virus infection

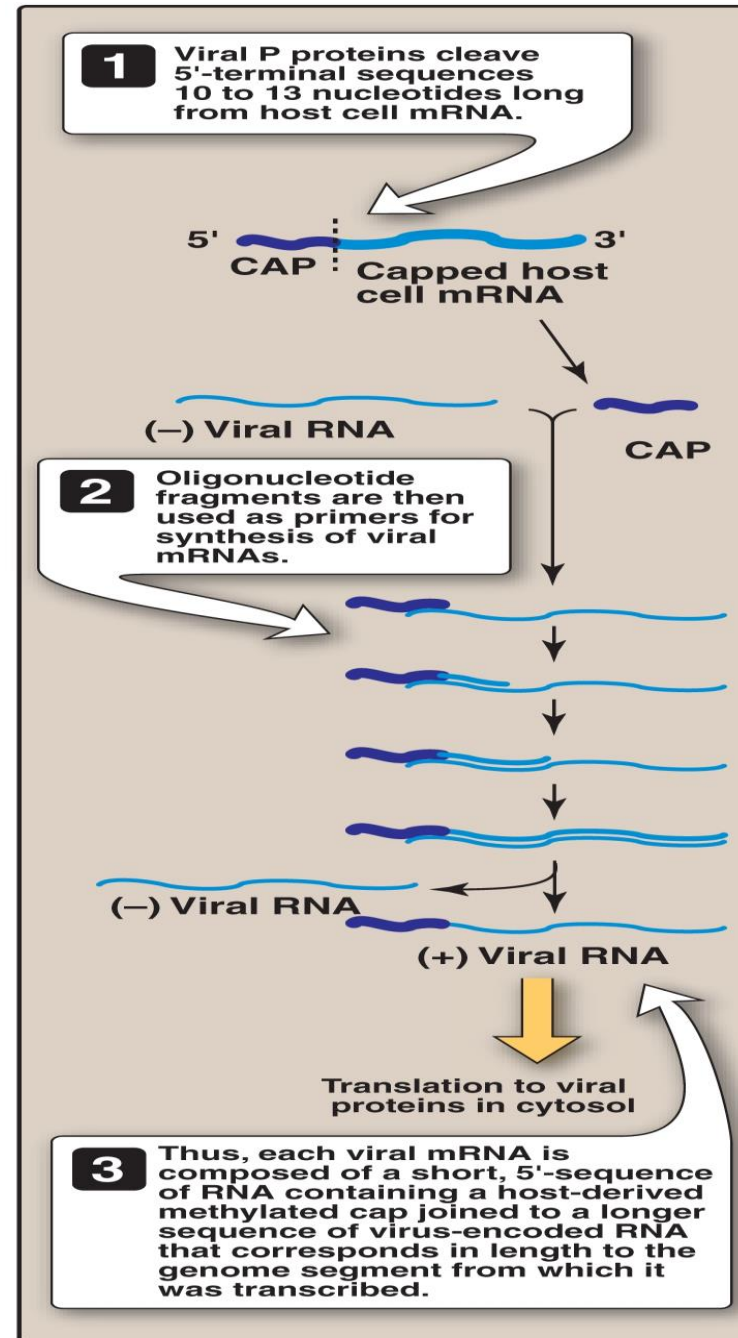
- What is special about the viral RNA sequence?



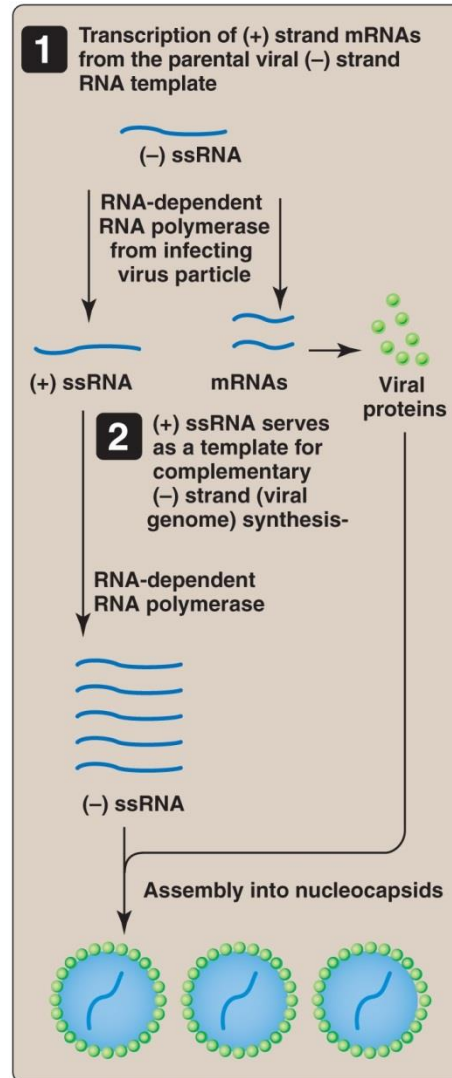
Capping



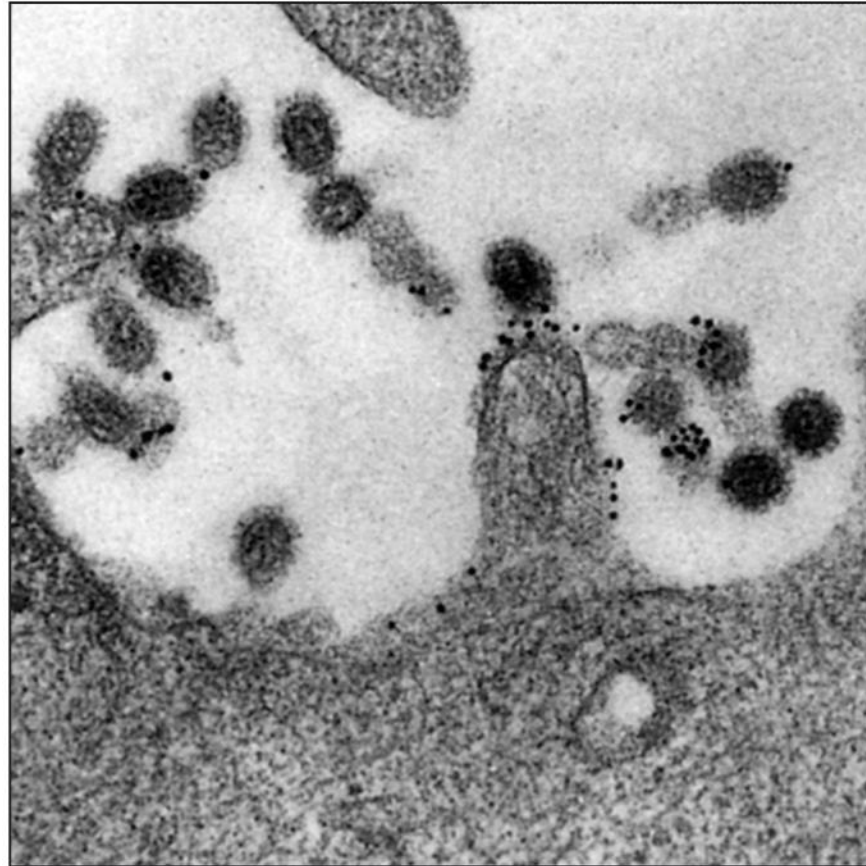
From – to + RNA



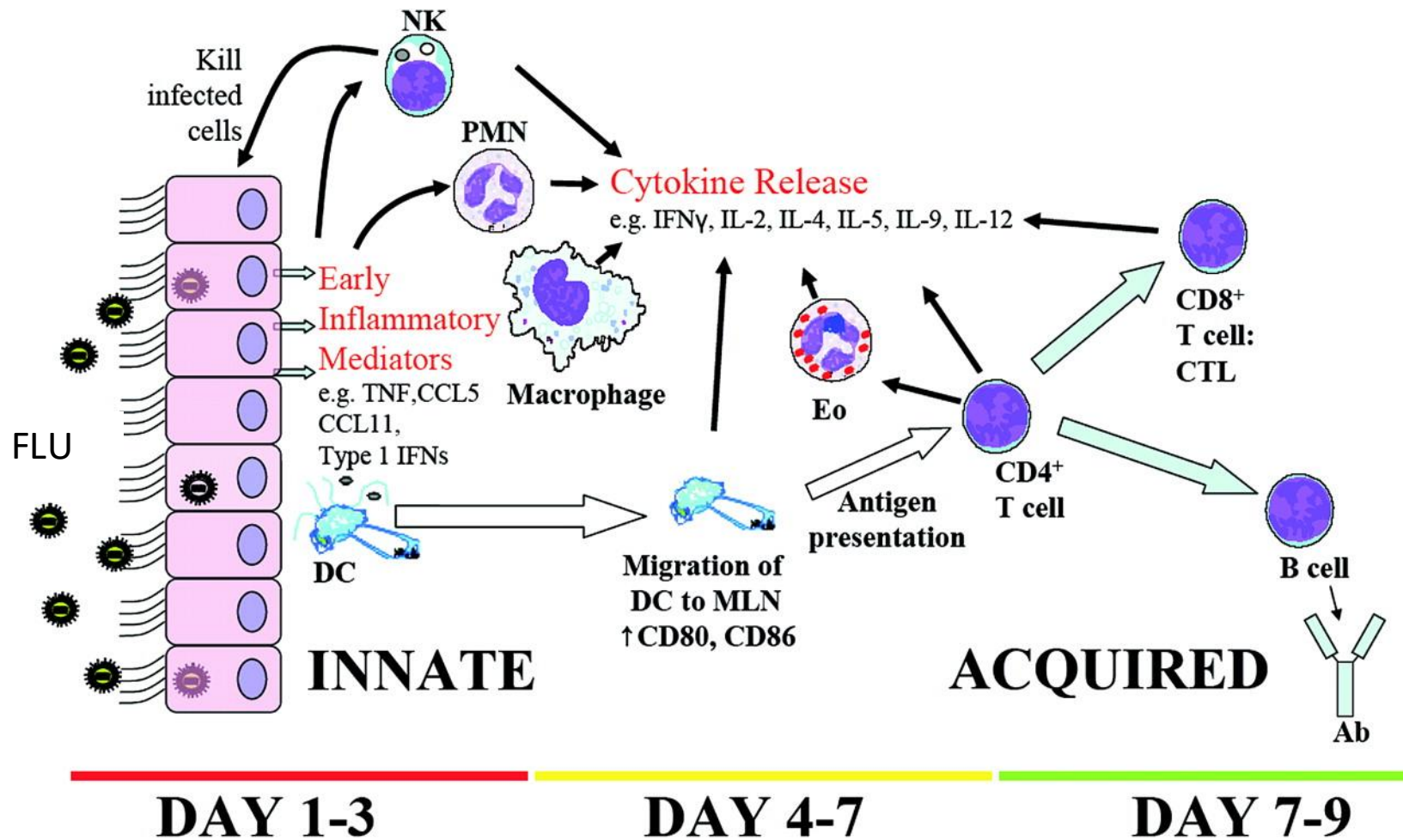
From + to - RNA



Virus release



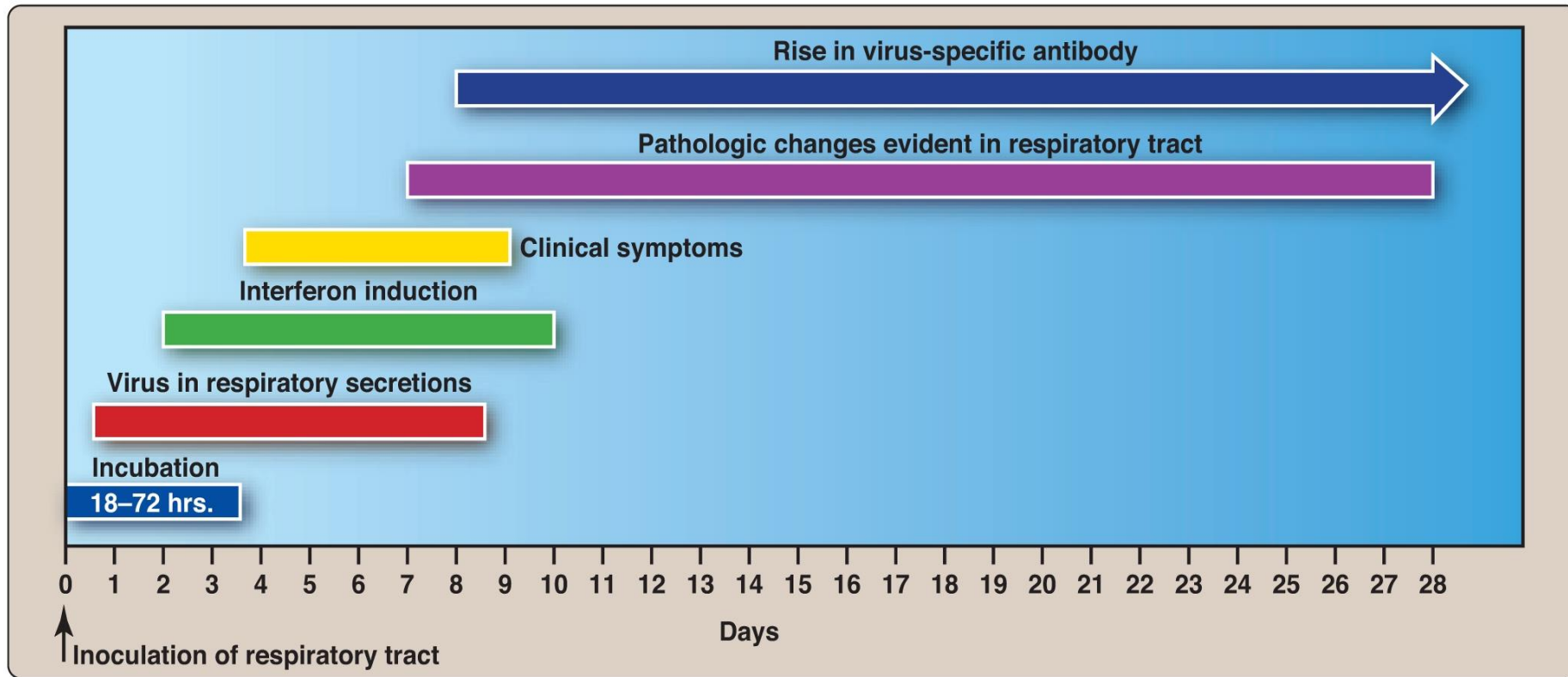
Cells involved in the immune response to FLU. Cellular infection triggers the release of early inflammatory mediators, e.g., TNF and IFN- α/β .



Peter J. M. Openshaw, and John S. Tregoning Clin.
Microbiol. Rev. 2005;18:541-555

Clinical Microbiology Reviews

Virus infection and disease



What will protect you from dying this 1st week?

Influenza type A, B, C and D

- There are 4 types of Influenza virus:
 - Type A can infect several species, birds, pigs, dogs humans
 - Type B can infect seals and humans
 - Type C human specific
 - Type D (Cattle and Pigs)
- (Para Influenza virus is not influenza, it is a different virus)
- <http://www.virology.ws/2009/09/22/the-a-b-and-c-of-influenza-virus/>

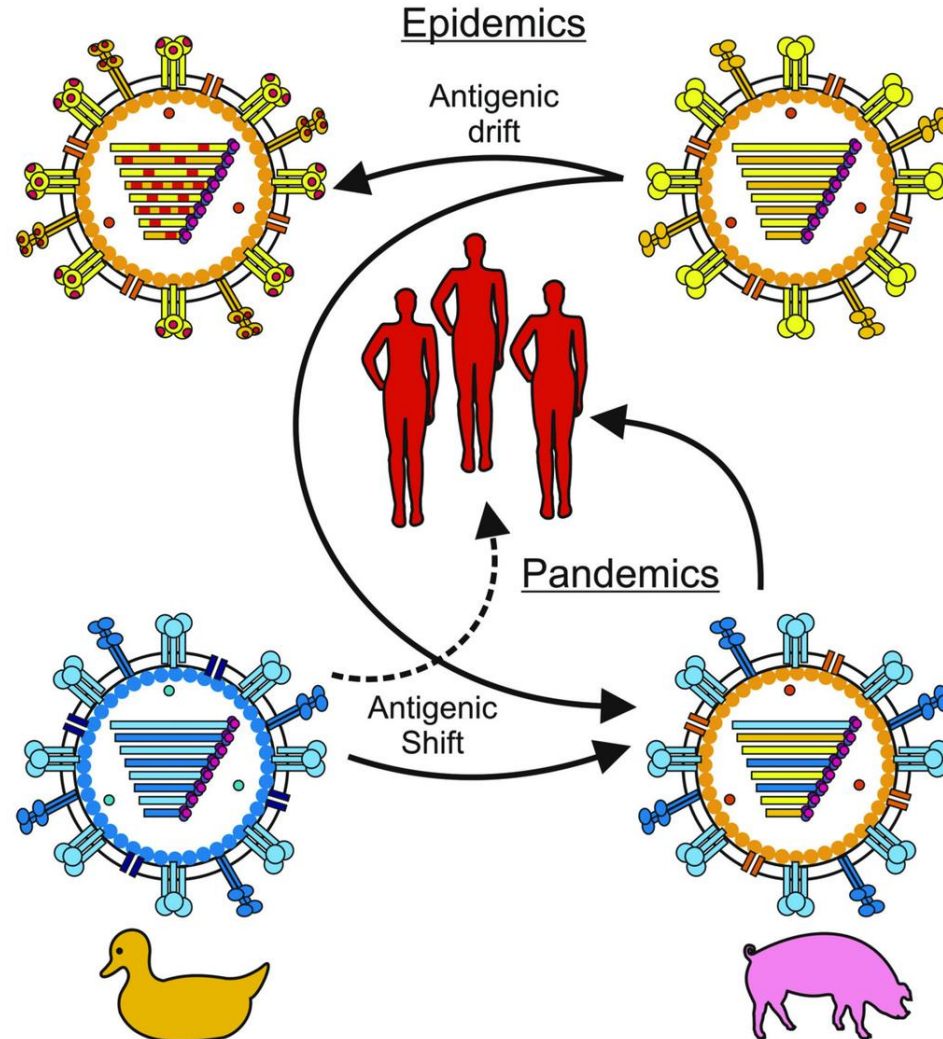
Influenza type A, B and C

- The Influenza virus vaccine holds Type A and B virus
- Type A can give pandemic, Type B can get people very sick
- In 2008 1/3 off all laboratory confirmed infected people died.

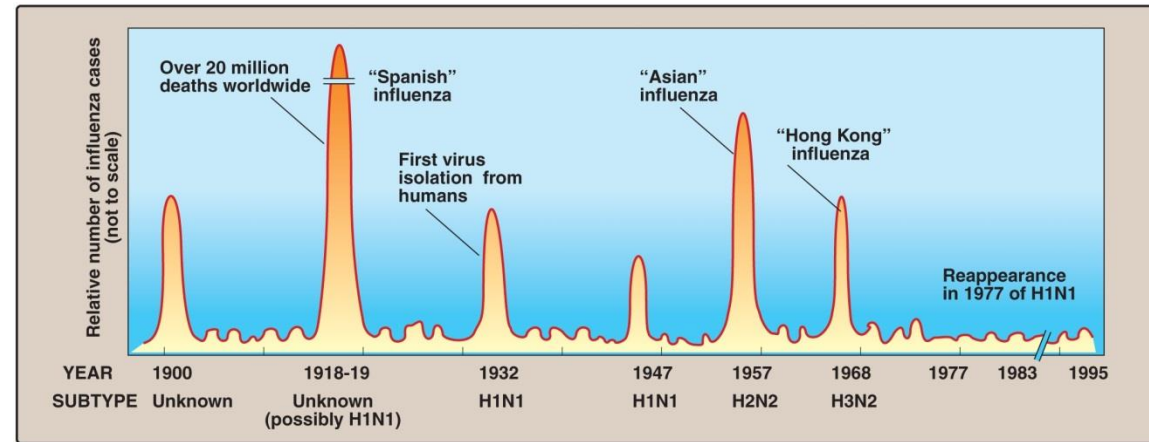
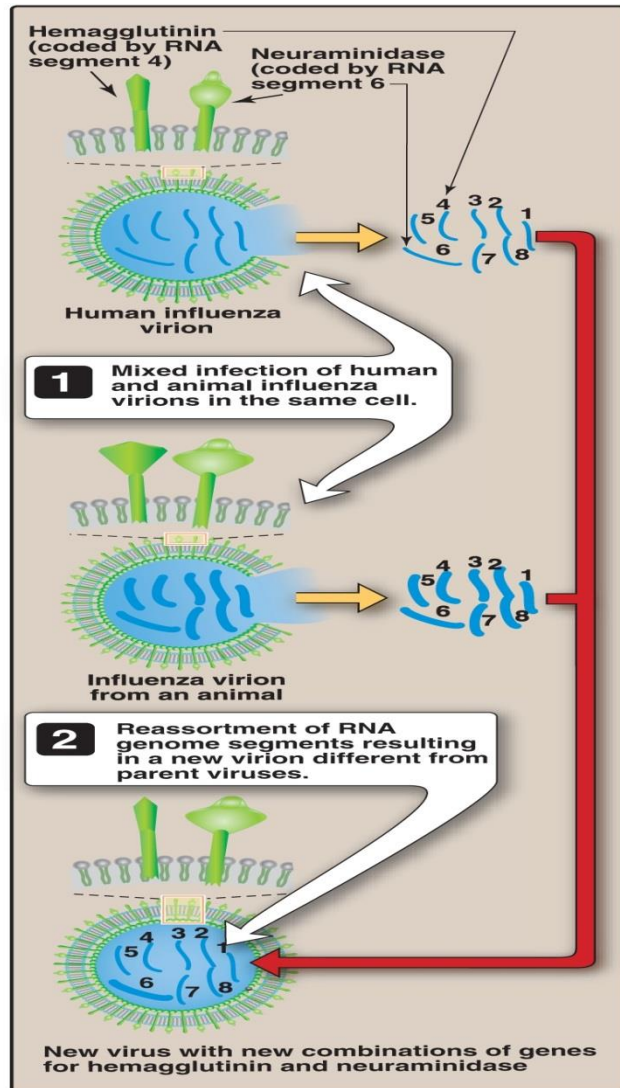
Antigenic drift/shift

- What is the difference between drift and shift?
- How can drift happen?
- How can shift happen?

Antigenic Shift and Drift

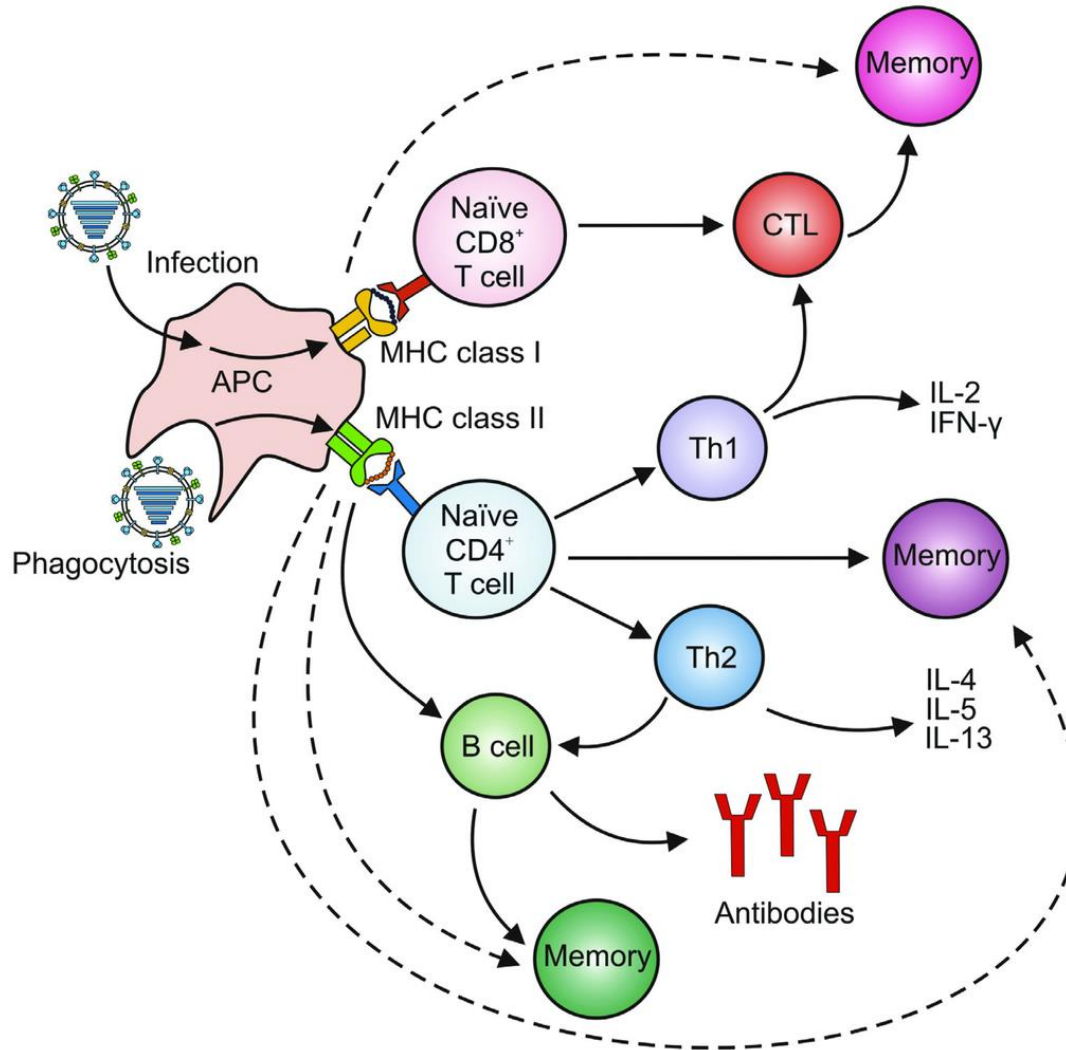


Antigenic drift/shift

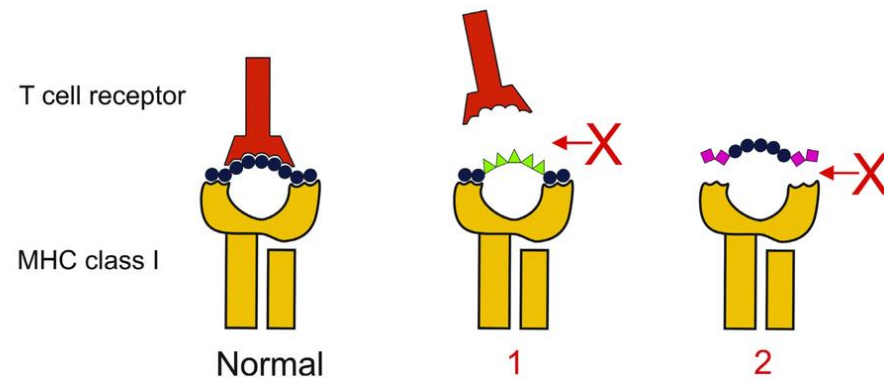
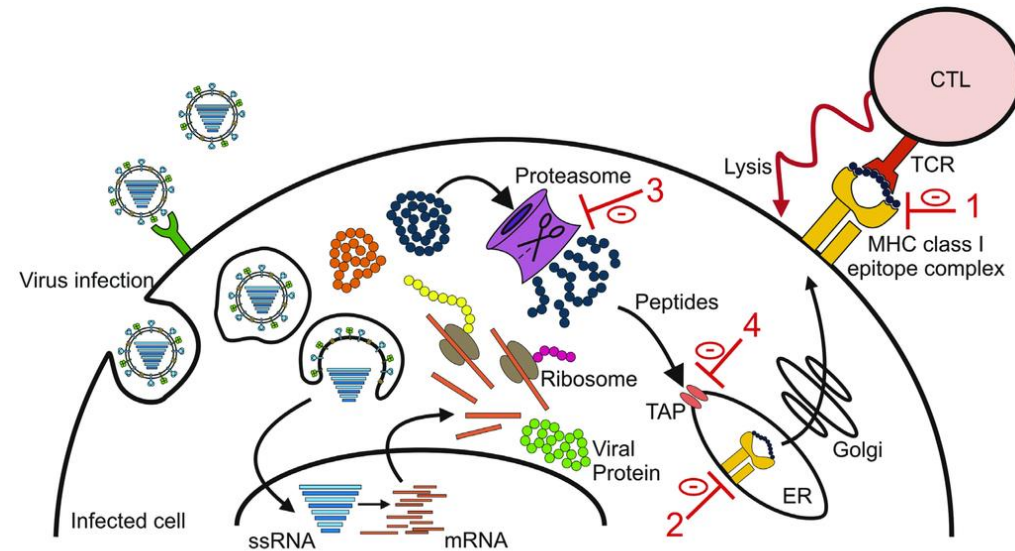


Copyright © 2013 Wolters Kluwer Health | Lippincott Williams & Wilkins

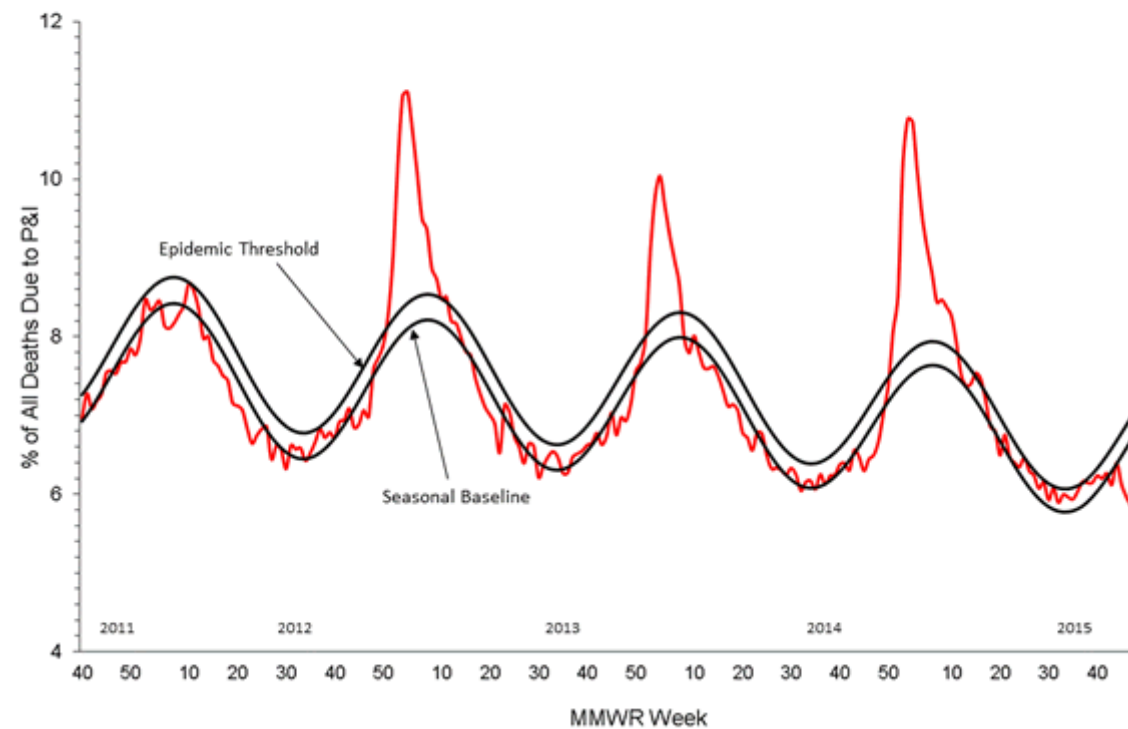
Immune response



Evasion of T cell response

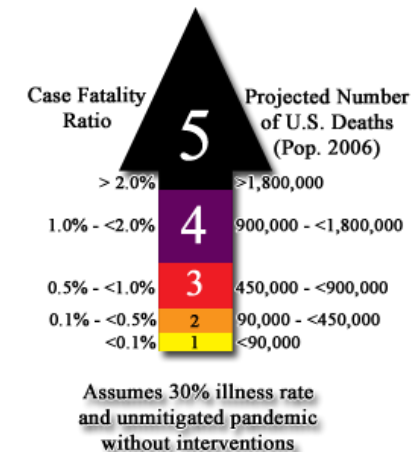


Pneumonia and Influenza Mortality from
the National Center for Health Statistics Mortality Surveillance System
Data through the week ending November 28, 2015, as of December 17, 2015



Influenza Pandemics

Known flu pandemics					
Name of pandemic	Date	Deaths	Case fatality rate	Subtype involved	Pandemic Severity Index
1889–1890 flu pandemic (Asiatic or Russian Flu)	1889–1890	1 million	0.15%	possibly H3N8 or H2N2	N/A
1918 flu pandemic (Spanish flu)	1918–1920	20 to 100 million	2%	H1N1	5
Asian Flu	1957–1958	1 to 1.5 million	0.13%	H2N2	2
Hong Kong Flu	1968–1969	0.75 to 1 million	<0.1%	H3N2	2
Russian flu	1977–1978	no accurate count	N/A	H1N1	N/A
2009 flu pandemic	2009–2010	105,700-395,600	0.03%	H1N1	N/A



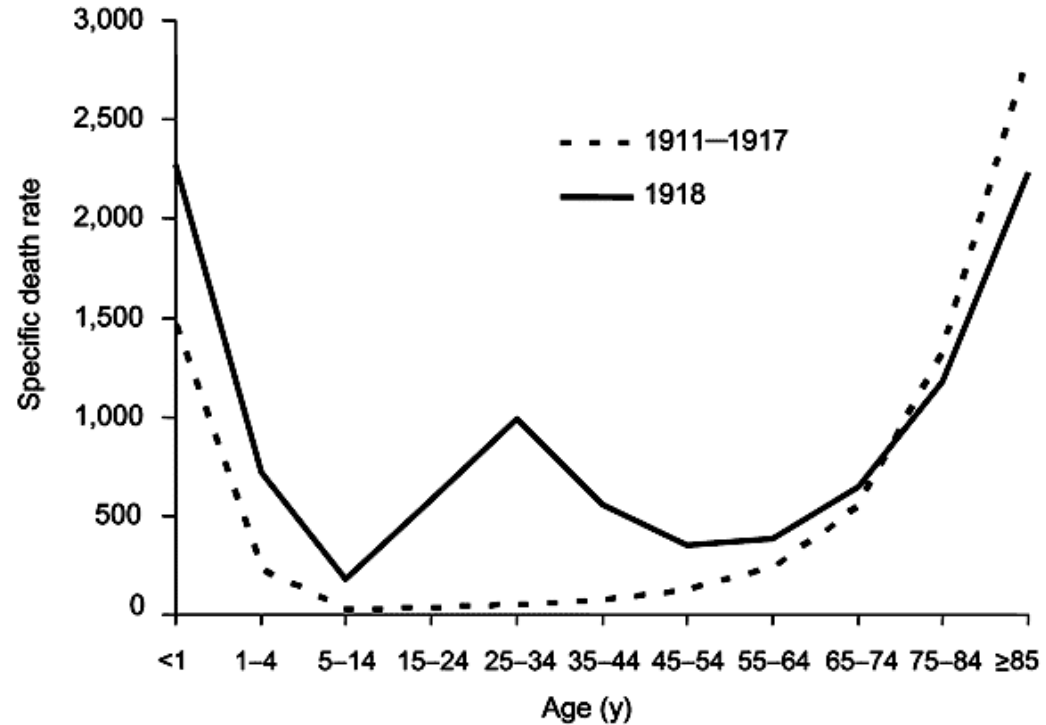
Influenza Virus

Table 2. Overview of genomic loci reported to be involved in enhanced mammalian pathogenicity of highly pathogenic Asian lineage H5N1 viruses			
Gene, Protein	Mutation	Effects	Reference
HA	polybasic endo-proteolytic cleavage site	advantage for systemic dissemination and replication (poultry, mammals)	various
NA	19-25 aa deletion in stalk region	adaptation to growth in chickens and turkeys (?)	Matrosovich 1999, Giannecchini 2006
PB2	627K	enhanced systemic replication in mice	Hatta 2001, Shinya 2004
	701N	increased pathogenicity in mice	Li 2005
PB-1	13P, 678N	enhanced polymerase activity; advantageous for early species-specific adaptation processes?	Gabriel 2005
NP	319K		
NS-1	92E	facilitated escape of innate immune responses, reduced viral clearance in pigs	Seo 2004

<http://www.influenzareport.com/ir/ai.htm>

Courtesy of Timm Harder

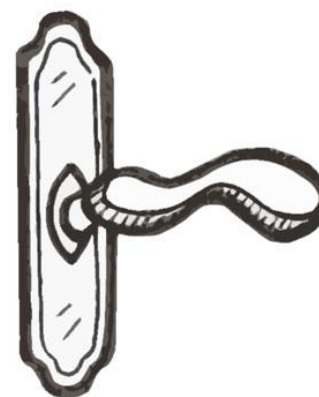
Spanish Flu Mortality



Why is there a dip in the ages of around 5-14 and 35 – 65 years old?

Spreading of the virus

- How does the virus spread?



Symptoms

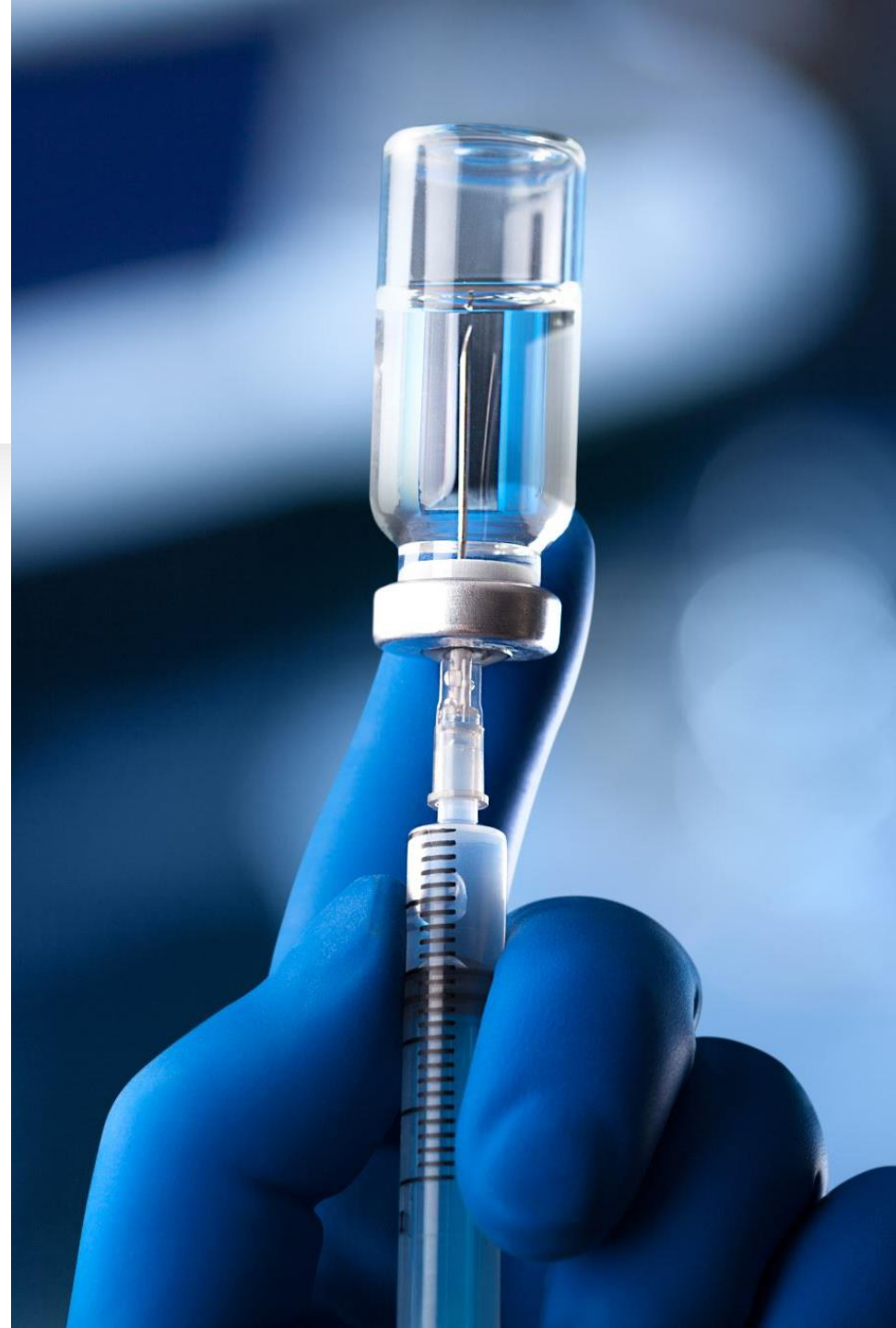
- Cough
- Throat problems
- Fever
- Muscle aches
- Drowsiness
- Headache

Symptoms

- Influenza is also a known trigger for a number of rarely encountered, yet often serious, CNS diseases, including
- the encephalopathy condition of Reye's syndrome,
- the peripheral neuropathy known as Guillain-Barre syndrome,
- the lesser known complaints of Kleine-Levin syndrome
- post-encephalitic Parkinson's disease.

Prevention

- Vaccine
 - Advantages and disadvantages?
 - How do the vaccines look like? (protein, DNA, RNA, Virus)
- **There are 3 recommended vaccines:**
- Fluzone High-Dose Quadrivalent vaccine.
- Flublok Quadrivalent recombinant flu vaccine.
- Fluad Quadrivalent adjuvanted flu vaccine.

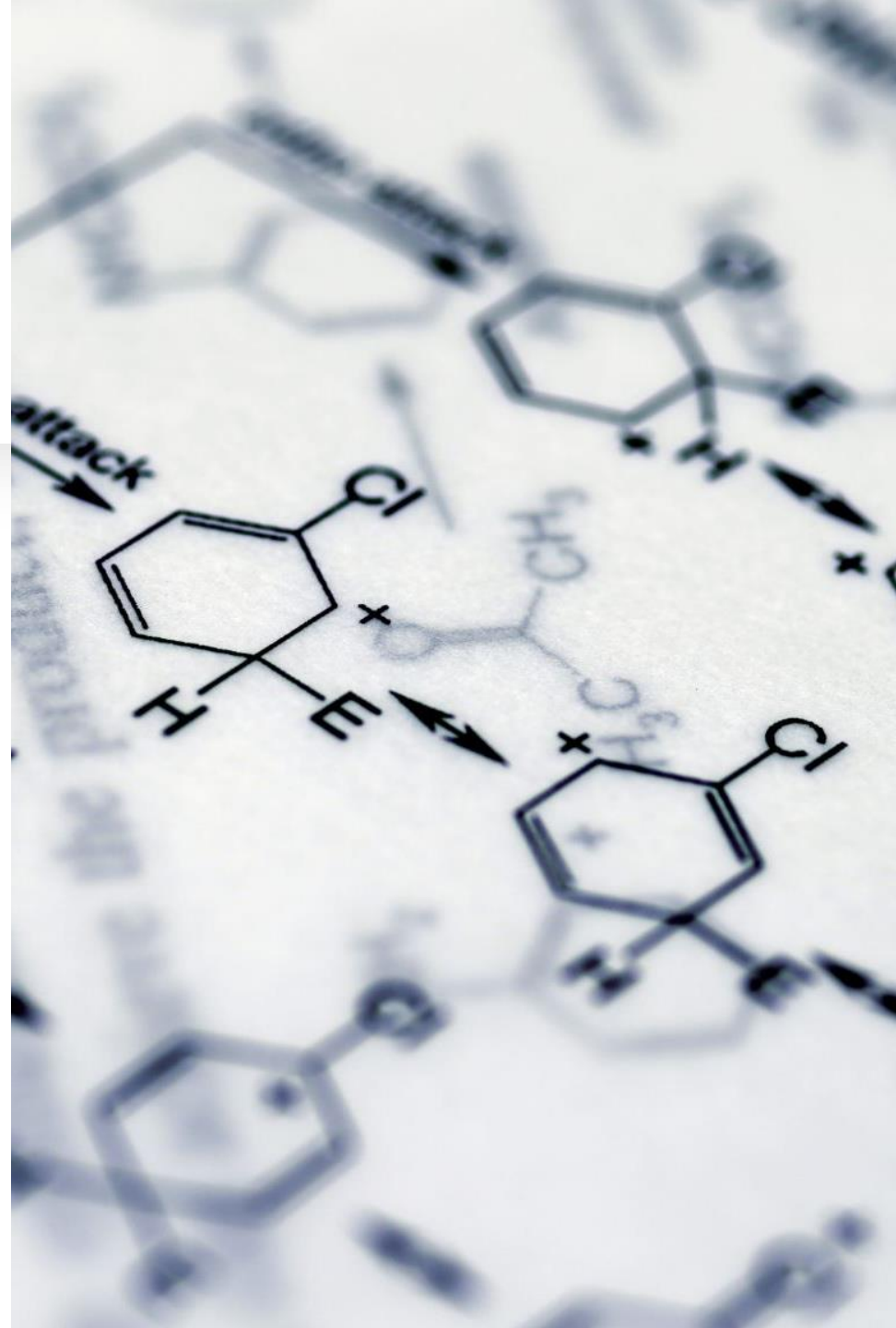


Prevention

- Alternative:
 - Amantadine (Block Viral M2)
 - Rimantadine (Block Viral M2)
 - Side effects (neurologic), and lack of efficacy for type B virus. Drug resistance.
 - Polymerase acidic endonuclease inhibitor
 - M2 is a proton channel involved in endosomal release of Virus
Type A M2 protein is different from type B

Treatment

- Zanamivir, Oseltamivir, Peramivir
(Neuraminidase inhibitor)
- Baloxavir (*CAP-dependent
endonuclease blocker*)
- Virus A and B



Treatment and prevention

- Prevention
- treatment
- How?

